

Accelerometer-Only Identification of Structural Parameters via Physics-Informed Evolutionary Artificial Neural Networks under Multi-Objective Optimization

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Abstract—This study presents a physics-guided Evolutionary Artificial Neural Network (EANN) framework for accelerometer-only identification of stiffness and damping parameters in civil structures, formulated as a multi-objective optimization problem. The proposed approach employs a two-phase evolutionary search strategy that integrates physical admissibility constraints directly into the neural mapping, decoupling the computationally intensive training phase from deployment and enabling structural parameter estimation via deterministic forward inference. A five-story building prototype, modeled as a multiple mass-spring-damper system, serves as the experimental benchmark under seismic base excitation. The EANN framework is rigorously compared against a multi-objective Genetic Algorithm (GA) and a Particle Swarm Optimization (PSO) baseline over 30 independent stochastic runs per method. Statistical validation through Kruskal–Wallis, post-hoc Wilcoxon rank-sum tests with Holm correction, nonparametric bootstrap confidence intervals, and Pareto dominance analysis confirms the robustness and repeatability of the proposed framework. Beyond accuracy, the EANN exhibits substantially reduced parametric dispersion, particularly in damping coefficients ($CV \approx 0.07$ – 0.10 across all floors), and generates a denser, more uniformly distributed Pareto front compared with GA and PSO. From a control-oriented perspective, reduced parametric variability translates directly into tighter eigenvalue uncertainty bounds on the identified state matrix, enhancing reliability for structural health monitoring, observer design, vibration control, and seismic engineering applications. In terms of median reconstruction accuracy across independent runs, the EANN achieves a reduction of 9.8% in median Mean Square Error (MSE) and 5.0% in median Root Mean Square Error (RMSE) relative to the PSO baseline, as derived from the median RMSE values reported across 30 independent runs.

Index Terms—Structural Parameter Identification, Neural Network, Genetic Algorithm, Particle Swarm Optimization, Seismic Engineering, Multi-Objective Optimization, Structural Health Monitoring, Accelerometer-Based Identification

Manuscript received XXX XX, 2026; revised XXX XX, 2026.

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I. INTRODUCTION

Modeling and identification of structural parameters are fundamental to control-oriented civil engineering applications, including structural health monitoring, active, semi-active, and passive vibration control for seismic response mitigation, and state estimation. In numerous studies, structural deformations are effectively represented using lumped mass–spring–damper models [1], [2], which are also widely used in damage detection [3] and observer-based state reconstruction [4].

In multi-story shear-building representations, stiffness and damping coefficients directly determine the eigenstructure of the system matrix and, consequently, the dynamic response characteristics. Accurate estimation of these parameters is therefore essential for reliable modal analysis, robust controller synthesis, and observer design. For example, state estimation using PI algebraic observers critically depends on proper parameter calibration [5]. Moreover, these models are embedded in modern seismic design regulations, including those of Mexico City [6], because they balance physical interpretability and computational tractability. Throughout this work, each floor mass is considered concentrated and modeled as a rigid diaphragm, while ground acceleration is recorded at the structure's base. The linear mass–spring–damper formulation enables efficient simulation and facilitates compatibility with optimization algorithms aimed at minimizing discrepancies between measured and simulated structural responses. This balance between model simplicity and descriptive accuracy makes lumped-parameter models particularly attractive for control applications.

Recent research has framed structural parameter identification as a multi-objective optimization problem. Rodríguez-Torres et al. [7] compared classical metaheuristics (GA, PSO, NSGA-II) with quantum-inspired variants for identifying stiffness and damping in multi-story structures, reporting improved convergence properties for quantum-inspired methods. However, their formulation relies on displacement and velocity measurements, which are difficult to obtain during real seismic events since any external reference frame moves with the ground motion.

This limitation motivates acceleration-only identification strategies, where non-inertial accelerometers are the primary sensing devices. Accelerometers are widely available in vibration-control systems [8], [9] and structural health monitoring [10]. Nonetheless, the inverse mapping from

acceleration measurements to structural parameters is highly nonlinear and sensitive, particularly when only limited sensing is available. Hybrid strategies combining classical experiment-design principles with neural networks have shown promise for real-time identification under limited sensing conditions in related engineering domains [11].

Neural networks have increasingly been applied to structural identification and monitoring tasks because they can learn complex input-output mappings directly from measured data without requiring an explicit analytical model of the underlying dynamics [12]. Early demonstrations of this capability employed one-dimensional convolutional neural networks (1D-CNNs) to extract damage-sensitive features directly from raw acceleration signals in real time, circumventing the need for handcrafted feature engineering that had previously limited the scalability of vibration-based detection systems [13].

For time-dependent structural responses under seismic excitation, recurrent architectures have shown particular promise. Deep long short-term memory (LSTM) networks can capture the temporal dependencies inherent in nonlinear building responses, enabling data-driven prediction of both elastic and inelastic behaviour without requiring physics-based time integration [14]. A further step toward bridging data-driven flexibility with physical consistency was taken by physics-guided convolutional neural networks (PhyCNN), which embed the laws of structural dynamics directly into the network training objective, substantially reducing the amount of labelled seismic input-output data required to achieve reliable predictive accuracy [15].

The broader principle of embedding physical knowledge into neural network architectures has been formalised under the framework of physics-informed machine learning, which unifies data-driven and model-based approaches by enforcing governing equations as soft constraints within the loss function [16]. Applied to multi-degree-of-freedom structural systems, this philosophy has been implemented through physics-based recurrent architectures inspired by implicit numerical integration schemes, demonstrating that structural dynamics priors can substantially reduce the trainable variable space while maintaining high accuracy across long response trajectories [17]. The neuroevolutionary paradigm further extends this landscape by replacing gradient-based weight optimisation with population-based evolutionary search, enabling the simultaneous optimisation of network topology and weights for non-differentiable or multi-objective fitness criteria [18]. In the context of physics-informed structural identification, this approach is particularly relevant because the objective landscape is typically non-convex and multi-modal, making gradient-based solvers sensitive to initialisation [19].

Transfer learning has emerged as a practical strategy to reduce the data acquisition cost associated with supervised structural identification. By pre-training a 1D-CNN on numerically generated structural response data and then fine-tuning it on limited experimental observations, significant improvements in convergence speed and generalisation have been demonstrated across heterogeneous structural configurations [20].

Beyond civil structures, evolutionary optimization and neural networks have been successfully applied to nonlinear

dynamical systems. Genetic algorithms have been used for identifying hysteretic magnetorheological damper models directly from experimental data [21], [22], while convolutional neural networks have demonstrated robustness in predicting nonlinear force responses [23]. Similarly, Integrated Neural Networks (INN) have been proposed for direct identification of nonlinear continuous-time models in state-space form under multi-objective training [24]. These developments highlight the growing potential of learning-based models to capture complex dynamics when properly guided by physical constraints.

Classical identification techniques also provide strong theoretical foundations. Instrumental-variable methods combined with integral filtering have enabled acceleration-only estimation of stiffness-to-mass and damping-to-mass ratios [25]. Frequency-domain identification strategies incorporating structural constraints have been validated on actively controlled prototypes [26]. Infinite-dimensional PDE-based identification has also been addressed through gradient-descent formulations [27], while deterministic-stochastic subspace methods have been developed for bilinear MIMO systems [28].

Despite these advances, most acceleration-based identification strategies rely on iterative reoptimization. In metaheuristic frameworks such as GA and PSO, each new excitation scenario entails the resolution of a global optimization problem formally expressed in (1).

$$\hat{\theta} = \arg \min_{\theta} J(\theta), \quad (1)$$

where $\theta = [K_s, C_s]$ represents stiffness and damping parameters and $J(\theta)$ is a multi-objective cost based on acceleration discrepancies. While effective, this paradigm entails repeated stochastic exploration, computational burden, and limited suitability for repeated deployment within control loops.

This paper introduces an alternative inference-based identification paradigm. Rather than solving a new global optimization problem for each data set, a Physics-Guided Evolutionary Artificial Neural Network (EANN) is trained offline within a multi-objective framework that incorporates structural constraints. After training, parameter estimation is carried out via forward inference, as expressed in (2).

$$\hat{\theta} = f_{\text{EANN}}(\ddot{u}(t)), \quad (2)$$

where $\ddot{u}(t)$ denotes measured floor accelerations. This decouples the computationally intensive evolutionary training phase from deployment. After convergence, the model parameters are frozen and structural identification reduces to a deterministic mapping with negligible computational cost.

From a control perspective, this distinction is crucial. Reduced parametric dispersion implies improved consistency of the identified state matrix, smaller uncertainty in modal characteristics, and enhanced robustness for controller and observer design. Furthermore, eliminating repeated global optimization enhances scalability and facilitates integration into structural health monitoring and adaptive control frameworks.

The main contributions of this paper are:

- A control-oriented formulation of acceleration-only structural parameter identification for a five-story building model.
- A physics-guided multi-objective evolutionary neural network that enforces admissible stiffness and damping estimates.
- A rigorous comparison with GA and PSO baselines under 30 independent stochastic runs per method.
- A comprehensive statistical validation including nonparametric hypothesis testing, parametric dispersion analysis, Pareto dominance metrics, and hypervolume evaluation.
- A control-theoretic interpretation linking parametric variability to structural robustness and computational deployment feasibility.

By calibrating stiffness and damping parameters such that the mass–spring–damper model reproduces measured floor accelerations and modal characteristics, the proposed framework enables reliable structural representation using non-invasive sensing. This capability is essential for structural health monitoring, vibration mitigation, and seismic engineering, where accurate yet computationally efficient models are required for inference, prediction, and decision-making.

The remainder of the paper is organized as follows. Section II describes the experimental platform of the reduced-scale five-story building prototype. Section III formulates the structural parameter identification problem under acceleration-only sensing, including the structural model, state-space representation, and multi-objective cost formulation. Section IV presents the physics-guided multi-objective EANN framework, the baseline optimization algorithms used for comparison, the computational complexity analysis, and the inference-based deployment paradigm. Section V reports the experimental results and statistical validation, including nonparametric hypothesis testing, bootstrap reliability analysis, parametric robustness assessment, and multi-objective quality indicators. Sections VI and VII provide the discussion and conclusions, respectively. **Notation:** The superscript e denotes base-10 exponent notation and $rand$ represents a random number in $(0, 1)$. The following acronyms are used: EANN (Evolutionary Artificial Neural Network), PSO (Particle Swarm Optimization), GA (Genetic Algorithm), MSE (Mean Squared Error), RMSE (Root Mean Square Error), MAE (Mean Absolute Error), R^2 (coefficient of determination), M_s (structural masses), K_s (structural stiffness), and C_s (structural damping).

II. EXPERIMENTAL PLATFORM

A reduced-scale five-story building prototype with dimensions of $600 \times 500 \times 1800$ mm is used for parametric identification using experimental data collected from non-inertial accelerometers (ADXL203E, range $\pm 1.7g$). The accelerometers were calibrated for each floor and showed an average sensitivity of $0.168g/V$. Each floor consists of an aluminum plate with side adaptations for displacement measuring purposes via Micro-Epsilon OptoNCDT 1302-200 laser sensors (range 200 mm, average gain 25.11 mm/V) as shown in Figure 1. It is important to note that the displacement sensors were used only for validation, i.e., to compare measured displacements

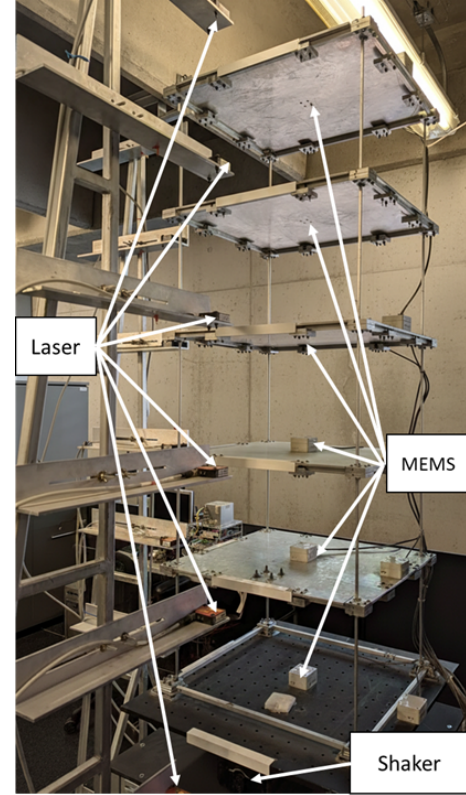


Fig. 1: Building Prototype

against model-based estimates. During assembly, all component masses were measured to compute the lumped mass of each floor; the corresponding values were 11.773, 9.17, 9.14, 9.12, and 9.08 kg. In addition, every level is supported by four aluminum columns with a height of 360 mm and a square cross-section of $1/4 \times 1/4$ in.

To reproduce earthquake-like base excitations, the prototype is mounted on a shaker driven by Parker servomotors (model 406T03LXR), with a maximum acceleration of 49.05 m/s^2 , a maximum velocity of 3 m/s , a position resolution of $5\text{ }\mu\text{m}$, and a travel range of 250 mm. Data acquisition and actuation are implemented in Simulink via two National Instruments PCI-6221-M boards, using a sampling time of 1 ms.

III. STRUCTURAL PARAMETER IDENTIFICATION UNDER ACCELERATION-ONLY SENSING

To address the challenge of parametric identification, it is essential to establish a mathematical framework that accurately characterizes the dynamic behavior of the system. This section presents the complete formulation adopted in this study, including structural modeling, modal analysis, and the multi-objective identification problem under physical constraints.

A. Structural Model and Dynamic Formulation

According to [29], the dynamic response of a civil structure subjected to unidirectional ground excitation can be mathematically described by the equation given in (3).

$$M_s \ddot{u}(t) + C_s \dot{u}(t) + K_s u(t) = -M_s l \ddot{u}_g(t). \quad (3)$$

Here, $\ddot{u}_g(t)$ denotes the ground acceleration induced by an earthquake. The relative displacements and their derivatives for each floor are defined as

$$\begin{aligned} u(t) &= [u_1(t), u_2(t), \dots, u_n(t)]^T \in \mathbb{R}^{n \times 1}, \\ \dot{u}(t) &= [\dot{u}_1(t), \dot{u}_2(t), \dots, \dot{u}_n(t)]^T \in \mathbb{R}^{n \times 1}, \\ \ddot{u}(t) &= [\ddot{u}_1(t), \ddot{u}_2(t), \dots, \ddot{u}_n(t)]^T \in \mathbb{R}^{n \times 1}, \end{aligned}$$

where n denotes the number of stories. The matrices $M_s, K_s, C_s \in \mathbb{R}^{n \times n}$ describe the mass, stiffness, and damping distributions of the structural system, respectively, whereas $l \in \mathbb{R}^{n \times 1}$ is the seismic influence vector governing the transmission of ground acceleration to each floor. Their explicit formulations are provided in (4)–(7).

$$M_s = \text{diag} [m_1 \quad m_2 \quad \dots \quad m_n] > 0, \quad (4)$$

$$K_s = \begin{bmatrix} k_1 + k_2 & -k_2 & \dots & 0 \\ -k_2 & k_2 + k_3 & \dots & \vdots \\ \vdots & \vdots & \ddots & -k_n \\ 0 & 0 & \dots & k_n \end{bmatrix} > 0, \quad (5)$$

$$C_s = \begin{bmatrix} c_1 + c_2 & -c_2 & \dots & 0 \\ -c_2 & c_2 + c_3 & \dots & \vdots \\ \vdots & \vdots & \ddots & -c_n \\ 0 & 0 & \dots & c_n \end{bmatrix} \geq 0, \quad (6)$$

$$l = [1 \quad 1 \quad \dots \quad 1]^T. \quad (7)$$

The parameters m_i, c_i, k_i denote the mass, damping, and stiffness of the i -th floor. In this study, structural masses are assumed known, and identification focuses exclusively on stiffness and damping parameters.

B. State-Space Representation and Modal Decomposition

For control-oriented modeling, the state representation should retain clear physical meaning while allowing a compact first-order formulation amenable to analysis and controller design. Thus, the state vector is constructed by stacking floor displacements and velocities, ensuring that each component is measurable or can be inferred from structural response measurements. This formulation is formally defined in (8).

$$x(t) = [u(t), \dot{u}(t)]^T \in \mathbb{R}^{2n \times 1}. \quad (8)$$

Consequently, the second-order dynamic system in (3) can be equivalently expressed in a state-space form, as defined in (9).

$$\begin{aligned} \dot{x}(t) &= Ax(t) + B\ddot{u}_g(t), \\ y(t) &= Cx(t), \end{aligned} \quad (9)$$

where the output matrix C establishes the mapping between the state vector and the measured structural responses. In the present formulation, it relates the states to the relative floor accelerations and is obtained as given in (10).

$$C = [-M_s^{-1}K_s \quad -M_s^{-1}C_s], \quad (10)$$

the matrices A and B characterize the state transition dynamics and the influence of ground acceleration on the structural velocity and displacement states, respectively. Their analytical expressions are provided in (11).

$$\begin{aligned} A &= \begin{bmatrix} 0_{n \times n} & I_{n \times n} \\ -M_s^{-1}K_s & -M_s^{-1}C_s \end{bmatrix}, \\ B &= \begin{bmatrix} 0_{n \times 1} \\ l \end{bmatrix}. \end{aligned} \quad (11)$$

It is worth noting that the eigenstructure of matrix A governs the natural frequencies and modal damping characteristics of the building, which highlights the importance of precise parameter identification for reliable control design. In addition, through modal decomposition, the floor displacements can be represented in the form provided in (12).

$$u(t) = \sum_{i=1}^n \phi_i q_i(t) = \Phi q(t), \quad (12)$$

where ϕ_i correspond to the modal shapes (eigenvectors) and $q_i(t)$ to the associated modal coordinates. Their dynamic behavior is governed by the natural frequencies ω_n , obtained from the generalized eigenvalue problem formulated in (13).

$$(K_s - \omega_n^2 M_s) \phi = 0. \quad (13)$$

In addition, ω_n can be experimentally extracted from acceleration data using Fourier Transform (FT) analysis by identifying dominant spectral peaks corresponding to resonant modes.

C. Acceleration-Based Multi-Objective Identification

Structural parameter identification can be posed as a multi-objective optimization problem in which an objective function aggregates the mismatch between measured and predicted relative accelerations according to several error criteria, while enforcing the physical plausibility of the structural model. The general formulation of this problem is given in (14).

$$\begin{aligned} \min_{\theta \in \mathcal{X}} F(\theta) &= (f_1(\theta), f_2(\theta), f_3(\theta)), \\ \text{subject to } g_i(\theta) &\leq 0, \quad i = 1, \dots, p, \\ \theta_{\ell_b} &\leq \theta \leq \theta_{u_b}. \end{aligned} \quad (14)$$

In this framework, the parameter vector θ gathers the structural stiffness (k_i) and damping (c_i) coefficients corresponding to the i th story, whose explicit formulation is provided in (15).

$$\theta = [k_1, \dots, k_n, c_1, \dots, c_n]^T; \quad (15)$$

The structural masses (m_i) are considered known a priori. To guarantee physically meaningful solutions, inequality constraints $g_i(\theta) \leq 0$ are imposed based on admissible material and construction properties, whose formulation is given in (16).

$$\begin{aligned} 0.7k_i - k_{i+1} &\leq 0 \\ -1.3k_i + k_{i+1} &\leq 0 \\ 0.7c_i - c_{i+1} &\leq 0 \\ -1.3c_i + c_{i+1} &\leq 0 \end{aligned} \quad \forall i \in 1, 2, \dots, n. \quad (16)$$

The stiffness (k_i) and damping (c_i) parameters are constrained by material-consistency requirements, enforcing $k_i > 0$ and $c_i \geq 0 \forall i$, which correspond to physically admissible positive stiffness and non-negative energy dissipation. Further inequality constraints arise from the assumption that structural columns are fabricated from the same material. Consequently, the proposed multi-objective formulation employs displacement and velocity states within the structural simulator for model-based acceleration reconstruction. Natural-frequency estimation (through FT and the modal eigenvalue problem in (13)), and the nearly monotonic variation imposed on k_i and c_i , as defined in (17).

$$\begin{aligned} f_1(\theta) &= \sum_{j=1}^n \frac{1}{N} \sum_{i=1}^N \left[\eta(\hat{u}_i(\theta) - \ddot{u}_i)^2 \right] \\ f_2(\theta) &= \sum_{j=1}^n (\hat{\omega}_{n_j}(\theta) - \omega_{n_j})^2 \\ f_3(\theta) &= \lambda_{\text{lin}} \left\| (A_{\text{ineq}} \hat{\theta} - b_{\text{ineq}})_+ \right\|_2^2 \\ &\quad + \lambda_{\text{sm}} \left(\sum_{i=1}^{n-1} (\hat{k}_{i+1} - \hat{k}_i)^2 + \sum_{i=1}^{n-1} (\hat{c}_{i+1} - \hat{c}_i)^2 \right) \\ &\quad + \lambda_{\text{log}} \left\| \log(\hat{\theta}_+) - \log(\theta_{\text{seed}}) \right\|_2^2, \end{aligned} \quad (17)$$

where $\ddot{u}_i$ and ω_{n_j} denote the measured accelerations and natural frequencies obtained via Fourier transform, while $\hat{u}_i(\theta)$ and $\hat{\omega}_{n_j}$ denote the accelerations and frequencies estimated by the model in (9). The data set comprises N samples from the accelerometer measurements; $(\cdot)_+ = \max(\cdot, 0)$ is applied element-wise; and $\hat{\theta}_+ = \max(\hat{\theta}, \epsilon)$. In addition, the linear constraint $A_{\text{ineq}} \hat{\theta} \leq b_{\text{ineq}}$ encodes near-monotonic k_i profiles typically observed in shear-building models. This formulation integrates acceleration consistency, modal alignment, and physically admissible structural patterns. Consequently, the identification problem balances data fidelity with structural realism, ensuring compatibility with control-oriented modeling requirements. This combination improves robustness in the identification process and reduces the risk of numerically accurate but physically unrealistic solutions.

IV. PHYSICS-GUIDED MULTI-OBJECTIVE EANN FRAMEWORK

This section presents the complete identification framework. The structural problem defined in Section III is reformulated as a physics-guided learning task, solved through a multi-objective evolutionary strategy. The section also includes the baseline reoptimization methods used for comparison.

A. Data and Feature Construction

Given the recorded base excitation signal $\{\ddot{u}_b(t_k)\}_{k=1}^N$ and the measured floor accelerations signals collected in the auxiliary multi-sensor array $\mathcal{A}_n \in \mathbb{R}^{N \times n}$, a fixed-length feature vector is constructed to summarize the most informative temporal and statistical content of the measured responses. In particular, for each floor $j = 1, \dots, n$ the last L samples of the j -th acceleration record are retained, and four global

descriptors are appended: root mean square, signal energy, kurtosis, and skewness. Let $\mathcal{A}_n \in \mathbb{R}^{N \times n}$ denote the j th column of $\mathcal{A}_n$, which represents the measured and filtered acceleration signal. For each channel, the feature-extraction map is defined as given in (18).

$$\phi_L(a_j) = \begin{bmatrix} a_j(N-L+1) \\ a_j(N-L+2) \\ \vdots \\ a_j(N) \\ \text{RMS}(a_j) \\ E(a_j) \\ \text{Kurt}(a_j) \\ \text{Skew}(a_j) \end{bmatrix} \in \mathbb{R}^{L+4}, \quad (18)$$

where the statistical descriptors are defined as given in (19).

$$\text{RMS}(a_j) = \sqrt{\frac{1}{N} \sum_{k=1}^N a_j^2(t_k)}, \quad E(a_j) = \sum_{k=1}^N a_j^2(t_k), \quad (19)$$

and $\text{Kurt}(a_j)$ and $\text{Skew}(a_j)$ denote the kurtosis and skewness of the complete signal, respectively. [30]

The full feature vector is then obtained by concatenating the contributions of all floors, as given in (20).

$$x_f = \Phi_L(\mathcal{A}_n) = \begin{bmatrix} \phi_L(a_1) \\ \phi_L(a_2) \\ \vdots \\ \phi_L(a_n) \end{bmatrix} \in \mathbb{R}^D. \quad (20)$$

where $\Phi_L(\cdot)$ denotes the feature extraction operator parameterized by the lag length L . Consequently, the dimensionality of the resulting feature vector is given in (21).

$$D = n(L+4). \quad (21)$$

The overall preprocessing and feature-construction pipeline for the n -story structure is illustrated in Fig. 2.

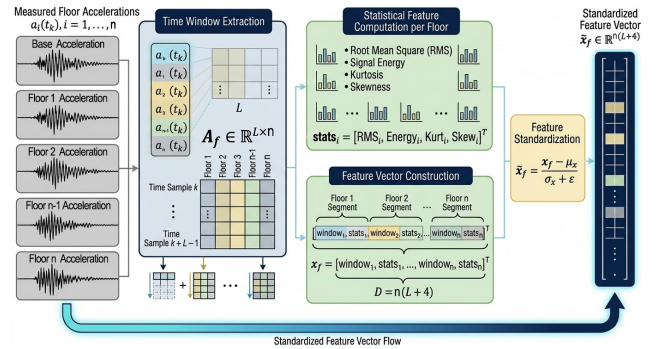


Fig. 2: Preprocessing and feature-construction pipeline for the n -story structural response signals.

Finally, the feature vector is standardized as given in (22).

$$\tilde{x}_f = \frac{x_f - \mu_x}{\sigma_x + \varepsilon}, \quad (22)$$

where μ_x and σ_x denote the mean and standard deviation of the feature vector, respectively, and ε is a small constant introduced for numerical stability.

B. EANN Mapping from Features to Physical Parameters

A compact multilayer perceptron (MLP) with two hidden layers is used to map the standardized feature vector $\tilde{x}_f$ to the physical parameters of the structural model. The nonlinear transformations are defined as given in (23)–(24).

$$z_1 = \tanh(W_1 \tilde{x}_f + b_1), \quad (23)$$

$$z_2 = \tanh(W_2 z_1 + b_2). \quad (24)$$

From this latent representation, two auxiliary outputs are generated, as defined in (25)–(26).

$$z_{\min} = \tanh(W_3 z_2 + b_3), \quad (25)$$

$$z_{\text{res}} = \tanh\left(W_3^{(r)} z_2 + b_3^{(r)}\right), \quad (26)$$

where $z_{\min}$ provides a coarse estimate of the parameters and z_{res} introduces a bounded residual correction.

The intermediate output is first mapped to the admissible physical range defined by the lower and upper bounds $[\ell_b, u_b]$, as given in (27).

$$\theta_{\min} = \ell_b + \frac{1}{2}(u_b - \ell_b)(z_{\min} + 1). \quad (27)$$

The final parameter vector is computed through a bounded residual adjustment mechanism, as defined in (28).

$$\hat{\theta} = \text{clip}(\theta_{\min} + 0.15 g_{\min} \odot x_{\text{res}}, \ell_b, u_b), \quad (28)$$

where $\odot$ denotes element-wise multiplication and $\text{clip}(\cdot)$ enforces the parameter bounds.

This formulation guarantees the physical admissibility of the estimated stiffness and damping parameters, while embedding prior physical constraints into the learning process, in line with recent developments in neuroevolution and physics-informed neural networks [18], [19]. The resulting vector $\hat{\theta} \in \mathbb{R}^{2n}$, obtained via the EANN mapping defined in Eqs. (27)–(28) is used as input to the structural simulator described in Algorithm 1 (function SIMULATE).

C. Two-Phase Multi-Objective Evolutionary Search

The parameter identification problem is solved using a two-phase genetic algorithm (GA) designed to balance global exploration and local exploitation of the search space.

a) Phase I: Exploration: An initial population of chromosomes is generated by randomly initializing the MLP parameters. Each chromosome encodes a candidate parameter vector w , which is MLP weights/biases mapped to the physical parameters θ through the EANN mapping.

To promote diversity and broad coverage of the search space, Phase I uses tournament selection, scattered crossover, and Gaussian mutation. The fitness of each chromosome is evaluated through the multi-objective function defined in Eq. (14), and Pareto dominance is used to identify non-dominated solutions [31], following established principles in evolutionary multi-objective optimization [32].

b) Phase II: Exploitation: The Pareto-optimal solutions obtained in Phase I are used to initialize a second evolutionary stage aimed at refining the search. In this phase, crossover remains the primary operator while mutation variance is reduced, allowing the algorithm to concentrate the search around promising regions of the parameter space.

Each chromosome is evaluated through the following steps:

- 1) decode $w \rightarrow \hat{\theta}$ via the EANN mapping,
- 2) simulate the structural model using $\hat{\theta}$,
- 3) compute the objective vector $F(w)$.

Parallel evaluation of chromosomes is used to accelerate computation.

c) Knee selection and local refinement: From the aggregated Pareto set $\{F(w_j)\}$, the final candidate is selected using the minimum Euclidean distance to the normalized utopia point, as defined in (29).

$$w^* = \arg \min_{w_j} \left\| \frac{F(w_j) - F_{\min}}{F_{\max} - F_{\min} + \varepsilon} \right\|_2. \quad (29)$$

Optionally, the selected solution may be further refined through a scalarized local optimization procedure, as given in (30).

$$\min_w \sum_{q=1}^3 w_q F_q(w), \quad w_q \propto \frac{1}{F_{\max,q} - F_{\min,q} + \varepsilon}. \quad (30)$$

The final structural parameter estimates are expressed as defined in (31).

$$\hat{\theta} = \theta(\tilde{x}_f; w^*). \quad (31)$$

D. Baseline Optimization Algorithms

For benchmarking purposes, two widely used evolutionary optimizers are considered: Particle Swarm Optimization (PSO) and a multi-objective Genetic Algorithm (GA). These methods serve as reference approaches for comparison with the proposed two-phase evolutionary search.

a) Particle Swarm Optimization: Particle Swarm Optimization (PSO) evolves a population of particles that explore the parameter space by combining inertial motion with attraction toward personal and global best solutions. The velocity and position update equations are defined as given in (32).

$$\begin{aligned} v_i(t_{k+1}) &= w_o v_i(t_k) \\ &\quad + c_1 r_1 (p_{\text{best},i} - x_i(t_k)) \\ &\quad + c_2 r_2 (g_{\text{best}} - x_i(t_k)), \\ x_i(t_{k+1}) &= x_i(t_k) + v_i(t_{k+1}). \end{aligned} \quad (32)$$

where x_i and v_i denote the position and velocity of particle i , $p_{\text{best},i}$ is the best position previously visited by particle i , and g_{best} is the best position found by the swarm. The random variables $r_1, r_2 \sim U(0, 1)$ introduce stochastic exploration. The inertia weight is gradually reduced according to the update rule given in (33).

$$w_o(t_{k+1}) = \alpha_w w_o(t_k), \quad (33)$$

which promotes exploration in early iterations and exploitation in later stages, as reflected in the hyperparameter configuration reported in Table I and supported by [33].

TABLE I: PSO Hyperparameters

Parameter	Value
Population size	100
Initial inertia weight w_0	0.8
Cognitive coefficient c_1	2
Social coefficient c_2	2
Inertia decay factor α_w	0.25

b) Genetic Algorithm: A multi-objective Genetic Algorithm (GA) is implemented to solve the constrained identification problem defined in (14). Each chromosome directly encodes the structural parameter vector $\theta = [k_1, \dots, k_n, c_1, \dots, c_n]^\top$ defined in (15). The evolutionary process is driven by Pareto dominance and crowding-distance ranking to maintain diversity across the population, with hyperparameters summarized in Table II.

At each generation, the algorithm performs:

- 1) tournament selection to choose parent chromosomes,
- 2) intermediate crossover to generate offspring,
- 3) Gaussian mutation to introduce stochastic perturbations,
- 4) Pareto-based survival selection to retain non-dominated solutions [34].

TABLE II: Genetic Algorithm Hyperparameters

Parameter	Value
Population size	220
Tournament size	6
Crossover fraction	0.70
Mutation operator	Gaussian (0.20, 0.95)
Pareto fraction	0.35
Maximum generations	160
Stall limit	$0.6 \times \text{MaxGen}$

E. Computational Complexity and Deployment Perspective

This subsection analyzes the computational burden of the proposed physics-guided EANN framework and contrasts it with classical reoptimization-based identification methods.

1) Training Complexity: Let d_θ denote the number of trainable parameters of the neural mapping, P the population size of the evolutionary algorithm, G the total number of generations (Phase I + Phase II), and C_{sim} the cost of propagating.

The dominant computational cost associated with the evolutionary training process can be approximated as expressed in (34).

$$C_{\text{train}} \approx P \cdot G \cdot C_{\text{sim}}. \quad (34)$$

Each chromosome evaluation requires decoding the neural parameters $w \mapsto \hat{\theta}$, propagating the structural model, and computing the objective vector $F(\theta) = [f_1, f_2, f_3]$. Consequently, the training process is computationally demanding but is performed only once in an offline learning stage.

The computational cost of the simulation scales with the state dimension n_s and the time horizon N of the acceleration signals, as defined in (35).

$$C_{\text{sim}} = O(n_s N), \quad (35)$$

since the discrete-time model must be propagated over the full measurement window, as commonly analyzed in the computational complexity of evolutionary algorithms [35].

2) Inference Complexity: After training, the structural parameters are inferred via a forward pass through the neural mapping, as defined in Eq. (36), thereby eliminating the need for iterative optimization during deployment.

$$\hat{\theta}^* = \hat{\theta}(\tilde{x}_f; w^*), \quad (36)$$

which corresponds to a sequence of matrix–vector operations across the MLP layers. Consequently, the inference computational complexity is expressed in (37).

$$C_{\text{infer}} = O(d_\theta), \quad (37)$$

which is independent of the population size, number of generations, or simulation horizon length.

In practice, this reduces the identification latency from seconds or minutes (per evolutionary optimization run) to milliseconds (per neural inference).

3) Comparison with Reoptimization-Based Methods: Classical identification approaches based on evolutionary optimization, such as GA or PSO, estimate the structural parameters g by directly minimizing the objective function $F(g)$ for each new dataset. Accordingly, the computational cost associated with each identification instance can be approximated as expressed in (38).

$$C_{\text{reopt}} \approx P \cdot G \cdot C_{\text{sim}}, \quad (38)$$

which must be incurred repeatedly whenever new measurements become available.

In contrast, the proposed framework converts the identification problem into a parametric regression task after a single offline optimization stage.

4) Complexity Comparison: Table III summarizes the computational characteristics of the considered approaches.

TABLE III: Computational complexity comparison between identification strategies [35]

Method	Training Cost	Online Cost	Deployment Suitability
PSO-based Identification	$P G C_{\text{sim}}$	$P G C_{\text{sim}}$	Limited
GA-based Identification	$P G C_{\text{sim}}$	$P G C_{\text{sim}}$	Limited
Proposed Physics-Guided EANN	$P G C_{\text{sim}}$ (offline)	$O(d_\theta)$	Real-time capable

5) Implications for Control-Oriented Deployment: From a control perspective, the distinction is fundamental. In structural monitoring or adaptive control scenarios, model parameters must often be updated as new acceleration measurements become available. Under reoptimization schemes, each update requires solving a high-dimensional multi-objective optimization problem.

In contrast, the trained EANN provides:

- instantaneous parameter estimation through forward evaluation,
- deterministic computational cost,

- feasibility guarantees via the bounded output mapping (28),
- embedded physical regularization through the objective term f_3 .

This shift from repeated online reoptimization to a single offline physics-informed learning stage constitutes a key methodological contribution of the proposed framework and enables near real-time structural parameter estimation suitable for digital twins, structural health monitoring, and adaptive control integration.

F. Algorithmic Summary

Algorithm 1 summarizes the proposed physics-guided EANN identification framework. The method combines feature extraction from auxiliary acceleration signals, neural decoding of structural parameters, and physics-based simulation of the structural dynamics. A two-phase evolutionary optimization strategy is used to explore the parameter space and subsequently refine the Pareto candidates to obtain the final structural parameter estimate. [36]

Algorithm 1. Two-Phase Physics-Guided EANN Identification

- 1: **Require:** $\mathcal{A}_n \in \mathbb{R}^{N \times n}$ (auxiliary multi-sensor array), $\mathbf{A}_r \in \mathbb{R}^{N \times n}$ (measured floor accelerations), $\ddot{u}_b \in \mathbb{R}^N$ (base excitation), $\mathbf{t} \in \mathbb{R}^N$ (time vector), $\mathbf{M}_s$ (structural masses), $[\ell_b, \mathbf{u}_b]$, $\boldsymbol{\theta}_{\text{seed}}$, G_1, G_2 (numbers of generations).
- 2: **Ensure:** Estimated structural parameter vector $\hat{\boldsymbol{\theta}}^* \in \mathbb{R}^{2n}$ and floor-wise performance metrics.
- 3: **procedure** PHYSICSGUIDEDEANN
- 4: $\mathbf{x}_f \leftarrow \text{BUILDFEATURES}(\mathcal{A}_n)$
- 5: $\tilde{\mathbf{x}}_f \leftarrow (\mathbf{x}_f - \boldsymbol{\mu}_x) \odot (\boldsymbol{\sigma}_x + \varepsilon)$
- 6: Define $\mathbf{F}(\mathbf{w}) = [F_1(\mathbf{w}), F_2(\mathbf{w}), F_3(\mathbf{w})]$ via EVALFITNESS
- 7: **Phase I — Exploration**
- 8: Initialize population $\mathbf{w} \in [-1, 1]^d$
- 9: Include seed chromosome $\mathbf{w}_{\text{seed}}$ such that

$$\text{DECODE}(\mathbf{w}_{\text{seed}}, \tilde{\mathbf{x}}_f, [\ell_b, \mathbf{u}_b]) \approx \boldsymbol{\theta}_{\text{seed}}$$

- 10: Evaluate $\mathbf{F}(\mathbf{w})$ in parallel
- 11: Evolve population for G_1 generations using tournament selection, scattered crossover, and Gaussian mutation
- 12: Extract non-dominated set (W_I, F_I)
- 13: **Phase II — Exploitation**
- 14: Initialize new population from W_I and perturbed copies
- 15: Evaluate $\mathbf{F}(\mathbf{w})$ in parallel
- 16: Evolve population for G_2 generations using crossover and reduced-variance mutation
- 17: Extract non-dominated set (W_{II}, F_{II})
- 18: Aggregate Pareto candidates:
- 19: $W \leftarrow W_I \cup W_{II}$
- 20: $F \leftarrow F_I \cup F_{II}$
- 21: Select compromise solution by

$$(i_{\text{knee}}, F_{\text{knee}}^{(n)}, \mathbf{s}) \leftarrow \text{KNEESELECT}(F)$$

- 22: $\mathbf{w}^* \leftarrow W[i_{\text{knee}}]$
- 23: Optionally refine locally around $\mathbf{w}^*$ by solving

$$\mathbf{w}^* \leftarrow \arg \min_{\mathbf{w}} \boldsymbol{\omega}^\top \mathbf{F}(\mathbf{w}), \quad \boldsymbol{\omega} = \frac{1}{\mathbf{s} + \varepsilon}$$

- 24: Normalize $\boldsymbol{\omega}$
- 25: $\hat{\boldsymbol{\theta}}^* \leftarrow \text{DECODE}(\mathbf{w}^*, \tilde{\mathbf{x}}_f, [\ell_b, \mathbf{u}_b])$
- 26: $\mathbf{A}_{\text{rel}} \leftarrow \text{SIMULATE}(\hat{\boldsymbol{\theta}}^*, \ddot{u}_b, \mathbf{t}, \mathbf{M}_s)$
- 27: Compute MSE, RMSE, MAE, and R^2 for each floor by comparing $\mathbf{A}_{\text{rel}}$ against $\mathbf{A}_r$
- 28: **return** $\hat{\boldsymbol{\theta}}^*$ and floor-wise metrics

29: **end procedure**

30: **function** BUILDFEATURES($\mathcal{A}_n$)

- 31: Extract the last L lagged samples from each column of $\mathcal{A}_n$
- 32: Concatenate global statistics (RMS, energy, kurtosis, and skewness) for each floor
- 33: **return** $\mathbf{x}_f \in \mathbb{R}^{n(L+4)}$
- 34: **end function**

35: **function** DECODE($\mathbf{w}, \tilde{\mathbf{x}}_f, [\ell_b, \mathbf{u}_b]$)

- 36: Unpack $\mathbf{w}$ into $\{\mathbf{W}_1, \mathbf{b}_1, \mathbf{W}_2, \mathbf{b}_2, \mathbf{W}_3, \mathbf{b}_3, \mathbf{W}_3^{(r)}, \mathbf{b}_3^{(r)}\}$
- 37: $\mathbf{z}_1 \leftarrow \tanh(\mathbf{W}_1 \tilde{\mathbf{x}}_f + \mathbf{b}_1)$
- 38: $\mathbf{z}_2 \leftarrow \tanh(\mathbf{W}_2 \mathbf{z}_1 + \mathbf{b}_2)$
- 39: $\mathbf{z}_{\min} \leftarrow \tanh(\mathbf{W}_3 \mathbf{z}_2 + \mathbf{b}_3)$
- 40: $\mathbf{z}_{\text{res}} \leftarrow \tanh(\mathbf{W}_3^{(r)} \mathbf{z}_2 + \mathbf{b}_3^{(r)})$
- 41: $\boldsymbol{\theta}_{\min} \leftarrow \ell_b + \frac{1}{2}(\mathbf{u}_b - \ell_b) \odot (\mathbf{z}_{\min} + 1)$
- 42: $\Delta \boldsymbol{\theta} \leftarrow 0.15 \boldsymbol{\theta}_{\min} \odot \mathbf{z}_{\text{res}}$
- 43: $\boldsymbol{\theta} \leftarrow \text{clip}(\boldsymbol{\theta}_{\min} + \Delta \boldsymbol{\theta}, \ell_b, \mathbf{u}_b)$
- 44: **return** $\boldsymbol{\theta}$
- 45: **end function**

46: **function** EVALFITNESS($\mathbf{w}$)

- 47: $\boldsymbol{\theta} \leftarrow \text{DECODE}(\mathbf{w}, \tilde{\mathbf{x}}_f, [\ell_b, \mathbf{u}_b])$
- 48: $\mathbf{A}_{\text{rel}} \leftarrow \text{SIMULATE}(\boldsymbol{\theta}, \ddot{u}_b, \mathbf{t}, \mathbf{M}_s)$
- 49: Compute $F_1(\boldsymbol{\theta})$
- 50: Compute $F_2(\boldsymbol{\theta})$
- 51: Compute $F_3(\boldsymbol{\theta})$ from Eq. (17)
- 52: **return** $\mathbf{F}(\mathbf{w}) = [F_1(\boldsymbol{\theta}), F_2(\boldsymbol{\theta}), F_3(\boldsymbol{\theta})]$
- 53: **end function**

54: **function** SIMULATE($\boldsymbol{\theta}, \ddot{u}_b, \mathbf{t}, \mathbf{M}_s$)

- 55: Build stiffness and damping matrices $\mathbf{K}(\boldsymbol{\theta})$ and $\mathbf{C}(\boldsymbol{\theta})$
- 56: Form the continuous-time state-space model
- 57: Discretize the model via the Tustin transform
- 58: Propagate the discrete-time state equation over $\mathbf{t}$ using $\ddot{u}_b$
- 59: Extract simulated floor accelerations $\mathbf{A}_{\text{rel}} \in \mathbb{R}^{N \times n}$
- 60: **return** $\mathbf{A}_{\text{rel}}$
- 61: **end function**

62: **function** KNEESELECT(F)

- 63: Normalize objective vectors with respect to $F_{\min}$ and $F_{\max}$
- 64: Compute the Euclidean distance of each normalized point to the utopia point
- 65: Select the candidate with minimum normalized distance
- 66: **return** $i_{\text{knee}}, F_{\text{knee}}^{(n)}$, and scaling vector $\mathbf{s}$
- 67: **end function**

V. EXPERIMENTAL RESULTS AND STATISTICAL VALIDATION

This section presents the experimental results obtained with the proposed physics-guided multi-objective EANN framework using measurements from the structural prototype. The preprocessing steps applied to the measured signals are described first. The structural parameters are then identified from the acceleration data, and the reconstructed structural responses are compared with the experimental measurements to evaluate the accuracy of the model. Finally, a comparative analysis of the considered optimization algorithms is carried out, complemented with statistical validation to assess the robustness and significance of the obtained performance differences. The validation addresses three key aspects: (1) **dynamic fidelity**, i.e., the capability of each method to accurately reproduce measured floor accelerations under base excitation; (2) **parametric robustness**, i.e., the variability of the identified stiffness and damping coefficients across stochastic runs, which directly impacts model confidence and controller sensitivity; and (3) **multi-objective trade-off consistency**, i.e., the stability of the balance between time-domain tracking accuracy, modal coherence, and physics-based regularization.

A. Experimental Setup, Data Preprocessing, and Evaluation Protocol

Before the identification procedure, the measured signals are preprocessed to reduce noise and ensure numerical stability during parameter estimation. The preprocessing stage consists of three main steps: digital filtering, bias removal, and normalization of the measured signals.

First, the raw acceleration and displacement measurements are digitally filtered using a second-order low-pass filter to remove high-frequency noise introduced by the sensors and the data acquisition system. The filter is parameterized by a cutoff frequency ω_c and a quality factor $Q_c = 1/\sqrt{2}$. Based on the spectral analysis of signals obtained via Fourier transform, the cutoff frequency is set to 6 Hz for the laser displacement sensors and 20 Hz for the MEMS accelerometers. This filtering step attenuates high-frequency disturbances while preserving the dominant structural dynamics of the system.

After filtering, the signals are centered by subtracting their median value to mitigate bias introduced by sensor offsets. For a measured signal $y(t)$, the centered signal $\tilde{y}(t)$ is defined as given in (39).

$$\tilde{y}(t) = y(t) - \bar{y}, \quad (39)$$

where $\bar{y}$ denotes the temporal median of the signal. Finally, the signals are normalized to improve numerical conditioning during the identification process, as defined in (40).

$$y_n(t) = \frac{\tilde{y}(t) - \mu_y}{\sigma_y}, \quad (40)$$

where μ_y and σ_y denote the mean and standard deviation of the centered signal, respectively. After preprocessing, the filtered and normalized acceleration signals from the five-floor structure, described in Section II, are arranged into the data matrix, as defined in (41).

$$\mathbf{A}_5 = \begin{bmatrix} a_1(t_1) & a_2(t_1) & \dots & a_5(t_1) \\ a_1(t_2) & a_2(t_2) & \dots & a_5(t_2) \\ \vdots & \vdots & \ddots & \vdots \\ a_1(t_N) & a_2(t_N) & \dots & a_5(t_N) \end{bmatrix}, \quad (41)$$

where $a_i(t_k)$ denotes the relative filtered, centered acceleration measured at floor i at time instant t_k , referred to as $\ddot{u}_i$ in the model (3). This matrix constitutes the input data set used for the structural parameter identification procedure described in Section III.

The records are obtained by applying the North–South component of the ground-acceleration history from the 1985 Mexico City earthquake as the excitation input to the experimental structure, after scaling the excitation amplitude to match the prototype. This record provides sufficient richness to enable reliable parameter identification. Additionally, it is worth noting that the optimization data set is formed from five experimental trials using the same excitation, providing increased experimental support for the identification procedure.

A comparative analysis was conducted using three optimization approaches: Particle Swarm Optimization (PSO), Genetic Algorithms (GA), and the proposed physics-guided EANN framework. All methods were applied to the same

experimental data set and configured with the hyperparameters reported in Section IV to ensure a fair comparison. For each method, 30 independent runs were performed to assess the stability and consistency of the obtained solutions. Algorithm performance was evaluated using the metrics in (42), computational time to convergence, and variability of the solutions across runs, assessing not only identification accuracy but also computational efficiency and robustness [37].

B. Accuracy Assessment and Statistical Inference

Dynamic fidelity was evaluated through four acceleration-based performance indicators computed across all floors, as expressed in (42).

$$\mathcal{M} = \{\text{MSE}, \text{RMSE}, \text{MAE}, R^2\}, \quad (42)$$

where the primary indicators for control-relevant reconstruction are RMSE, MAE, and R^2 (scale-consistent and interpretable). Since normality was not guaranteed, nonparametric testing was adopted following recommended practice for comparing evolutionary and swarm-based algorithms [38].

a) *Global test (Kruskal–Wallis)*.: For each metric, the Kruskal–Wallis statistic is computed as defined in (43).

$$H = \frac{12}{N(N+1)} \sum_{i=1}^k n_i \bar{R}_i^2 - 3(N+1), \quad (43)$$

where $k = 3$ algorithms, $N = 90$ observations, $n_i = 30$, $df = k - 1 = 2$ degrees of freedom, and $\bar{R}_i$ is the average rank for group i . Table IV reports the resulting p -values, showing statistically significant differences across all metrics.

TABLE IV: Kruskal–Wallis global test for acceleration metrics (30 runs per algorithm).

Metric	p -value
RMSE _{mean}	5.71×10^{-6}
MAE _{mean}	9.75×10^{-6}
MSE _{mean}	1.22×10^{-15}
R^2_{mean}	1.10×10^{-6}

b) *Pairwise test (Wilcoxon rank-sum with Holm correction)*.: To identify which algorithm pairs differ, post-hoc rank-sum tests with Holm correction were used. Table V summarizes the corrected p -values for the main accuracy metrics, confirming significant differences among all pairs ($p < 0.001$ in all cases).

TABLE V: Post-hoc Wilcoxon tests with Holm correction (corrected p -values).

Metric	GA vs PSO	GA vs EANN	PSO vs EANN
RMSE _{mean}	2.69×10^{-4}	3.07×10^{-4}	3.07×10^{-4}
MAE _{mean}	2.69×10^{-4}	3.47×10^{-4}	3.47×10^{-4}
MSE _{mean}	8.77×10^{-7}	1.29×10^{-10}	8.77×10^{-7}
R^2_{mean}	2.69×10^{-4}	2.69×10^{-4}	2.69×10^{-4}

c) *Descriptive accuracy across runs*.: Beyond statistical significance, control-oriented structural identification requires evaluating the *stability* of the identified parameters across independent optimization runs. Table VI reports mean $\pm$ standard deviation and median [IQR] values for the main acceleration-based performance metrics. GA achieves competitive median

error levels in some indices but is characterized by substantially larger dispersion, as reflected by higher standard deviation and interquartile range values. In contrast, EANN maintains a comparable central tendency while consistently exhibiting reduced variability across runs, which is desirable for reliable deployment of identified models in subsequent control design stages. Although PSO attains stable mean performance, the near-constant metric values observed across runs suggest limited solution diversity and potential premature stagnation, rather than robust multi-objective exploration (see Subsec. V-C). Overall, the improved balance between accuracy and statistical repeatability provided by EANN supports its suitability for control-relevant structural model identification.

TABLE VI: Acceleration accuracy across 30 runs: mean $\pm$ std and median [IQR].

Metric	Algorithm	Value
RMSE _{mean}	GA	0.225 $\pm$ 0.120
		Median: 0.160 [0.160, 0.160]
	PSO	0.199 $\pm$ 0.000
		Median: 0.199 [0.199, 0.199]
	EANN	0.221 $\pm$ 0.076
		Median: 0.189 [0.187, 0.196]
MAE _{mean}	GA	0.163 $\pm$ 0.085
		Median: 0.117 [0.117, 0.117]
	PSO	0.135 $\pm$ 0.000
		Median: 0.135 [0.135, 0.135]
	EANN	0.161 $\pm$ 0.059
		Median: 0.137 [0.135, 0.142]
R^2_{mean}	GA	0.787 $\pm$ 0.214
		Median: 0.903 [0.903, 0.903]
	PSO	0.841 $\pm$ 0.000
		Median: 0.841 [0.841, 0.841]
	EANN	0.812 $\pm$ 0.122
		Median: 0.860 [0.850, 0.864]

d) *Bootstrap reliability (confidence intervals).*: To evaluate run-to-run reliability without assuming specific distributional properties, nonparametric bootstrap confidence intervals were computed [39] for the mean of each acceleration-based performance metric. The resulting intervals, summarized in Table VII, show that EANN consistently produces finite and comparatively narrow confidence bounds across the main indices, indicating improved statistical stability of the identified structural dynamics under stochastic optimization conditions. In contrast, GA exhibits noticeably wider bootstrap intervals, reflecting higher variability and reduced repeatability across independent runs. Although PSO achieves competitive mean accuracy values, the nearly degenerate confidence intervals observed for several metrics suggest limited solution diversity, consistent with premature convergence phenomena in multi-objective search processes. From a control-oriented identification perspective, the balanced trade-off between accuracy and statistical repeatability provided by EANN supports its suitability for reliable structural parameter estimation and subsequent controller tuning. Complete bootstrap statistics are provided as supplementary material.

C. Multi-Objective Quality and Control-Theoretic Analysis

The structural identification problem was cast as a three-objective optimization problem, as expressed in (44).

$$\min_{\theta} \mathbf{F}(\theta) = [f_1(\theta), f_2(\theta), f_3(\theta)], \quad (44)$$

TABLE VII: Nonparametric bootstrap confidence intervals (95%) for the mean acceleration-based performance metrics across 30 independent runs. MSE_a values are derived as the square of the corresponding median RMSE across runs and are reported for reference only; primary accuracy indicators are RMSE_a, MAE_a, and R^2_a . Note that MSE_a is not directly comparable to the MSE_{mean} reported in Tables IV and V, which is computed as the mean of per-run MSE values across the 30 independent runs.

Metric	Algorithm	Mean	CI Low	CI High
MSE _a	GA	2.56×10^{-2}	1.99×10^{-2}	3.48×10^{-2}
	PSO	3.96×10^{-2}	3.96×10^{-2}	3.96×10^{-2}
	EANN	3.57×10^{-2}	3.49×10^{-2}	3.88×10^{-2}
RMSE _a	GA	0.225	0.188	0.272
	PSO	0.199	0.199	0.199
	EANN	0.221	0.197	0.250
MAE _a	GA	0.163	0.137	0.196
	PSO	0.135	0.135	0.135
	EANN	0.161	0.143	0.184
R^2_a	GA	0.787	0.704	0.853
	PSO	0.841	0.841	0.841
	EANN	0.812	0.765	0.849

where f_1 represents weighted acceleration tracking accuracy, f_2 promotes modal consistency through dominant spectral peak alignment, and f_3 enforces physical plausibility via parameter regularization and smoothness constraints. This formulation aims to obtain identification models that are not only accurate but also dynamically consistent and suitable for control-oriented applications, consistent with classical criteria for assessing convergence and diversity in evolutionary multi-objective optimization [40].

a) *Dominance structure.*: Let P_A and P_B denote the Pareto sets obtained by algorithms A and B . The dominance count metric is formally defined in (45).

$$D(A \succ B) = \sum_{\mathbf{p} \in P_A} \sum_{\mathbf{q} \in P_B} \mathbb{I}(\mathbf{p} \preceq \mathbf{q}), \quad (45)$$

which quantifies how frequently solutions from A dominate those from B across the objective space. The results reveal a pronounced asymmetry: EANN dominates GA 351,294 times, whereas GA dominates EANN only 35,410 times. This indicates that the proposed approach generates a denser population of balanced solutions instead of achieving improvements only at isolated extreme regions of the search space.

b) *Diversity and global coverage.*: Uniform Pareto coverage is critical in control-oriented identification problems, as different operating conditions may emphasize time-domain accuracy, modal fidelity, or parameter regularity. Pareto front diversity is therefore quantified using the spread indicator, as expressed in (46).

$$\Delta = \frac{\sum_{i=1}^M |d_i - \bar{d}|}{(M-1)\bar{d}}, \quad (46)$$

where d_i denotes the Euclidean distance between consecutive normalized Pareto solutions. The hypervolume (HV) indicator was employed to quantify the dominated region of the objective space relative to a reference point, as expressed in (47).

$$HV = \int_{\Omega} \mathbb{I}(\exists \mathbf{p} \in P : \mathbf{p} \preceq \mathbf{z}) d\mathbf{z}. \quad (47)$$

Table VIII summarizes the multi-objective quality indicators computed across 30 independent runs. The hypervolume indicator was evaluated using a normalized reference point defined as the worst objective vector observed across all algorithms and runs. EANN achieves a perfect normalized hypervolume of 1.0000 ± 0.0000 across all 30 runs, indicating that its Pareto front consistently dominates the reference point in all three objectives without exception. GA attains a comparable hypervolume of $0.99999 \pm 6.45 \times 10^{-6}$, confirming broad global coverage of the objective space. PSO does not produce a distributed Pareto front, collapsing to a single solution per run; consequently, spread and hypervolume indicators are not applicable, consistent with the premature convergence behavior.

Although EANN and GA achieve nearly identical hypervolume values, this indicator captures dominated volume but does not reflect internal solution distribution or concentration. The complementary spread indicator reveals that EANN (1.2062 ± 0.0956) and GA (1.2132 ± 0.0764) achieve comparable coverage uniformity, while the larger front size of EANN (98 solutions) versus GA (77 solutions) indicates finer resolution of trade-off candidates. Critically, the dominance asymmetry analysis confirms that EANN provides a more favorable internal distribution of solutions across the objective space, which is not captured by the hypervolume alone.

TABLE VIII: Multi-objective quality indicators across 30 independent runs. HV computed with a normalized reference point defined as the worst objective vector observed across all algorithms. PSO does not produce a distributed Pareto front due to single-point convergence per run; spread and HV indicators are therefore not applicable (N/A).

Algorithm	Spread Mean $\pm$ Std	Front Size	HV Mean $\pm$ Std
GA	1.2132 ± 0.0764	77	$0.99999 \pm 6.45 \times 10^{-6}$
PSO	N/A	N/A	N/A
EANN	1.2062 ± 0.0956	98	1.0000 ± 0.0000

c) Control-theoretic implications of parametric dispersion.: The five-DOF structural dynamics are described by (3) and equivalently expressed as the state-space system in (9), where the matrices A and B depend on the identified structural parameters, compactly represented as expressed in (48).

$$\theta = [k_1, \dots, k_5, c_1, \dots, c_5]^\top. \quad (48)$$

Across stochastic realizations, θ can be treated as a random variable, thereby inducing model uncertainty, as expressed in (49).

$$A(\theta) = A(\bar{\theta}) + \Delta A, \quad \Delta A = A(\theta) - A(\bar{\theta}). \quad (49)$$

A smaller dispersion of θ implies smaller $\|\Delta A\|$, which is directly related to control robustness. For a diagonalizable matrix $A(\bar{\theta}) = V\Lambda V^{-1}$, the Bauer–Fike theorem establishes an upper bound on eigenvalue perturbations [41], as expressed in (50).

$$\forall \lambda(\theta) \in \sigma(A(\theta)), \exists \hat{\lambda} \in \sigma(A(\bar{\theta})) : |\lambda(\theta) - \hat{\lambda}| \leq \kappa(V) \|\Delta A\|, \quad (50)$$

where $\kappa(V) = \|V\| \|V^{-1}\|$. Therefore, methods yielding lower dispersion in k_i and c_i tend to produce tighter pole

uncertainty, which is a desirable property for robust control and observer tuning. From a control-oriented identification standpoint, the richer and more uniformly distributed Pareto structure achieved by EANN facilitates the selection of structural models that better balance prediction accuracy, modal consistency, and physical regularization requirements.

D. Parametric Robustness, Physical Plausibility, and Control Usability

A control-oriented identification framework must yield parameter estimates θ with low stochastic dispersion and physically consistent spatial variation along the structure. The robustness of the identified parameters is quantified using the coefficient of variation (CV) computed across 30 independent optimization runs, as expressed in (51).

$$CV(p) = \frac{\sigma(p)}{\mu(p)}, \quad p \in \{k_i, c_i\}. \quad (51)$$

a) Floor-wise variability of stiffness and damping.:

Table IX reports the resulting CV values. The proposed EANN framework yields consistently low variability in the damping coefficients across all floors (approximately 0.07–0.10), indicating stable identification of the energy dissipation properties of the structure. In contrast, GA exhibits substantially higher damping dispersion in intermediate floors, with CV values exceeding 2 and 4 for c_3 and c_4 , respectively. Such variability suggests sensitivity to stochastic initialization and reduced reliability of the identified dynamic behavior. PSO presents near-zero variability for all parameters; however, when interpreted together with the descriptive accuracy and bootstrap analyses, this behavior is consistent with premature convergence and insufficient exploration of the multi-objective search space rather than with meaningful physical robustness.

TABLE IX: Coefficient of variation (CV) across 30 runs for stiffness (k_i) and damping (c_i) parameters. Lower values indicate higher control reliability. [†]PSO values reflect single-point convergence per run (premature stagnation), not genuine parametric robustness; see Section V-C.

Algorithm	Floor	CV(k_i)	CV(c_i)
GA	1	0.031	0.071
	2	0.062	0.285
	3	0.051	2.568
	4	0.070	4.025
	5	0.113	0.025
PSO	1	$\approx 0^\dagger$	$\approx 0^\dagger$
	2	$\approx 0^\dagger$	$\approx 0^\dagger$
	3	$\approx 0^\dagger$	$\approx 0^\dagger$
	4	$\approx 0^\dagger$	$\approx 0^\dagger$
	5	$\approx 0^\dagger$	$\approx 0^\dagger$
EANN	1	0.050	0.090
	2	0.058	0.071
	3	0.043	0.098
	4	0.033	0.092
	5	0.046	0.093

b) Implications for control-oriented modeling.: Uncertainty in damping parameters has a direct impact on vibration attenuation capability and closed-loop stability margins. Considering the identified state matrix $A(\theta)$, reduced dispersion in θ leads to a smaller perturbation ΔA , which in turn tightens

the eigenvalue uncertainty bounds derived from the Bauer–Fike theorem (see (50)). From a practical control perspective, this translates into: (i) improved predictability of observer convergence dynamics, (ii) enhanced robustness margins against modeling mismatch, and (iii) increased reliability of model predictive control strategies for vibration mitigation. Under accelerometer-only sensing conditions, where parameter observability is inherently limited, such robustness can be more critical than marginal improvements in median tracking error, since controller performance may degrade significantly under parameter drift.

c) *Spatial coherence and physical smoothness.*: Beyond dispersion magnitude, structural parameters are expected to follow spatially coherent patterns along the building height. To evaluate physical plausibility, a normalized smoothness index is introduced, as expressed in (52).

$$S(p) = \frac{\sum_{i=1}^4 (p_{i+1} - p_i)^2}{\sum_{i=1}^5 p_i^2}, \quad (52)$$

where lower values indicate smoother and more physically consistent parameter distributions. Empirical analysis shows that EANN produces significantly smoother damping profiles compared with GA, reducing unrealistic floor-to-floor discontinuities and yielding a more coherent dynamic representation suitable for control design and structural health monitoring applications, which is consistent with recent evidence showing that physics-guided neural models improve robustness and physical interpretability in structural identification problems [42].

E. Visual Evidence and Reconstructed Structural Responses

In addition to the statistical validation presented in the previous subsections, visual analysis provides complementary insight into the stability, dispersion, and multi-objective convergence behavior of the identification algorithms. From a control-oriented perspective, graphical evidence is particularly useful to evaluate consistency across stochastic runs, spatial coherence of identified parameters, and the practical quality of Pareto trade-offs.

Figure 3 compares displacement time histories obtained experimentally with model predictions computed using the best-identified parameters. The close agreement between the measured data (red dots) and the predicted responses (blue lines) supports the accuracy of the estimated structural parameters and highlights the ability of the proposed methodology to capture the dynamic behavior of the structure.

Figure 4 shows the distribution of the main acceleration-based performance metrics across the 30 independent runs. The proposed EANN framework exhibits tighter clustering and reduced dispersion compared with GA and PSO. Although GA occasionally attains slightly lower metric values in isolated runs, its broader spread indicates reduced repeatability. PSO presents nearly constant values across runs, consistent with

early convergence rather than robust multi-objective exploration. The compact distributions obtained by EANN support the descriptive statistics reported in Table VI and indicate improved reliability for control-oriented structural modeling.

Figure 5 illustrates the scatter distribution of the identified stiffness (k_i) and damping (c_i) parameters across floors. EANN produces more concentrated parameter estimates with smoother spatial transitions between consecutive floors. In contrast, GA exhibits large variability, particularly in damping parameters of intermediate floors, which may introduce uncertainty in modal properties and controller tuning. PSO again shows minimal variability, consistent with stagnation behavior.

Figure 6 presents the Pareto fronts obtained in the normalized three-objective space. The EANN method generates a denser and more uniformly distributed set of non-dominated solutions, indicating improved convergence and better trade-off coverage between acceleration tracking accuracy, modal consistency, and physical regularization. GA tends to populate extreme regions of the objective space, while PSO provides limited Pareto diversity. From a control design viewpoint, the richer Pareto structure achieved by EANN facilitates the selection of structural models that balance prediction accuracy and robustness requirements.

Overall, the visual evidence complements the statistical validation results and reinforces the conclusion that EANN provides more consistent and control-usable structural models under stochastic optimization conditions.

F. Computational Cost and Deployment Perspective

Computational runtime is an important practical aspect in optimization-based structural identification. However, in control-oriented applications, runtime must be interpreted within the broader context of model usability and reliability. Structural parameter identification is typically performed offline, or at a supervisory time scale, with the primary objective of obtaining a dynamically consistent and robust model suitable for subsequent controller synthesis, observer design, or structural health monitoring.

Table X summarizes the average computational cost per stochastic run for the considered algorithms. PSO achieves the lowest runtime due to its rapid convergence behavior; nevertheless, the nearly deterministic parameter outcomes observed in previous analyses suggest premature stagnation and limited exploration of the multi-objective search space. GA exhibits moderate computational cost but with higher variability, reflecting sensitivity to stochastic initialization and reduced consistency across runs. The proposed EANN framework incurs a higher computational cost due to the integration of neural mapping and enhanced multi-objective optimization; nevertheless, it achieves superior parametric robustness, smoother damping profiles, and improved Pareto dominance. From a control-oriented perspective, this trade-off is acceptable since identification is performed offline, while controller performance critically depends on model fidelity.

VI. DISCUSSION

The experimental results reported in Section V establish that reliable structural identification under accelerometer-only

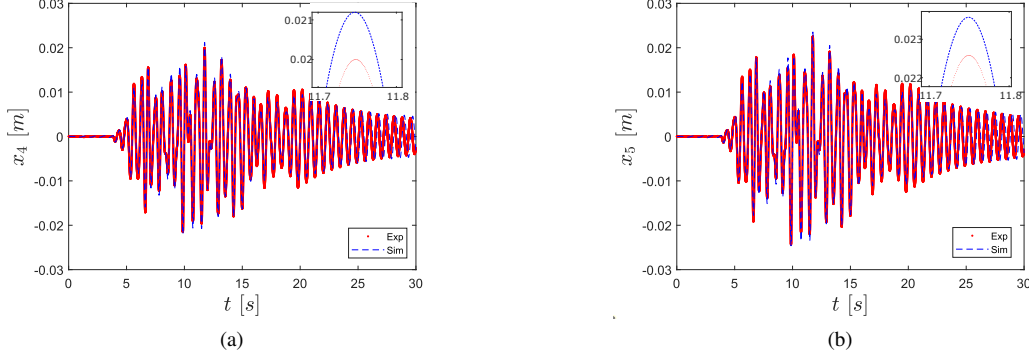
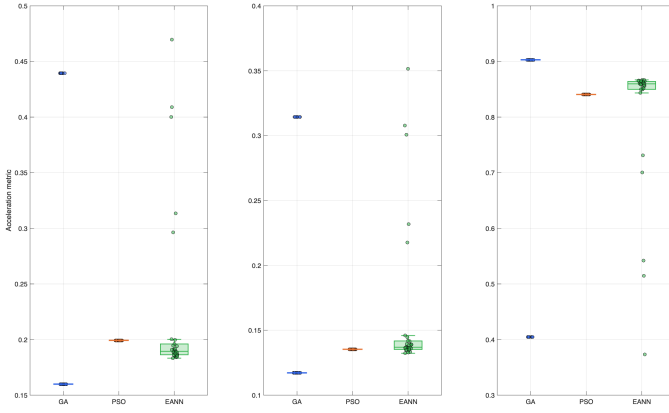
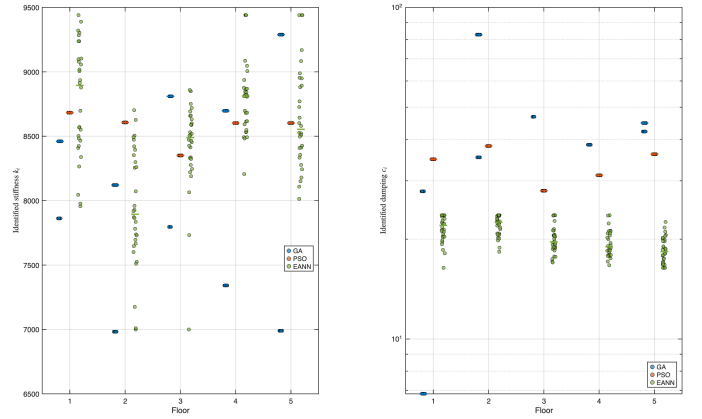
Fig. 3: Displacement comparison (a) x_4 . (b) x_5 .Fig. 4: Boxplot distribution of floor acceleration identification metrics (RMSE, MAE, and R^2) over 30 independent runs. EANN shows lower dispersion and tighter clustering around favorable performance levels than GA and PSO, indicating more consistent structural parameter identification.Fig. 5: Scatter distribution of identified stiffness (k_i) and damping (c_i) parameters across floors over 30 independent runs. EANN produces tighter parameter clustering and reduced dispersion compared with GA and PSO, indicating more stable floorwise identification.

TABLE X: Computational cost per stochastic run.

Algorithm	Runtime (s) mean $\pm$ std	Control interpretation
PSO	51.34 $\pm$ 6.82	Fast convergence but limited exploration of the multi-objective space.
GA	91.53 $\pm$ 14.93	Moderate cost with higher stochastic variability across runs.
EANN	438.75 $\pm$ 1.01	Higher computational cost with improved robustness and Pareto consistency.

sensing cannot be reduced to minimizing a single scalar error metric. The proposed EANN framework was evaluated along four dimensions — accuracy and statistical inference, multi-objective quality, parametric robustness, and computational deployment — each of which carries direct implications for control-oriented structural modeling. The following discussion synthesizes the evidence across these dimensions and situates the findings within the broader context of structural health monitoring and adaptive control design.

A. Accuracy and Statistical Inference

The nonparametric tests reported in Section V-B confirm that performance differences among GA, PSO, and EANN are statistically significant across all metrics ($p < 0.001$). However, statistical significance alone does not determine practical utility for control deployment.

GA occasionally achieves lower median error values in isolated runs, yet its substantially wider bootstrap confidence intervals and higher interquartile ranges confirm that such improvements are not reproducible across independent stochastic trials. From a control-oriented perspective, this variability directly propagates into model uncertainty during repeated identification sessions, which is particularly problematic in structural health monitoring applications where parameter estimation must remain consistent under varying excitation conditions.

PSO attains a consistent mean accuracy, but its near-degenerate confidence intervals reveal that this apparent consistency arises from premature convergence to a single point rather than from robust multi-objective exploration of the parameter space. Mechanistically, this behavior is consistent

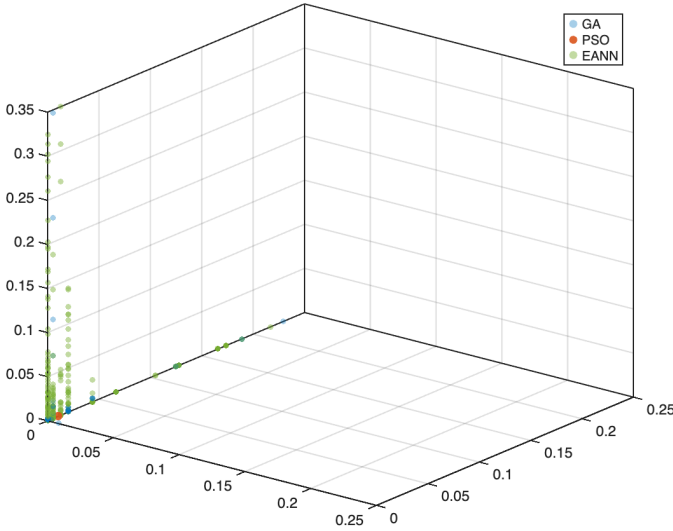


Fig. 6: Pareto fronts obtained in the normalized three-objective space over 30 independent runs. EANN generates a denser and more compact set of non-dominated solutions compared with GA and PSO, indicating improved convergence and trade-off distribution in acceleration-based structural identification.

with the well-documented velocity collapse phenomenon in PSO [33]: as the inertia weight decays according to (33), individual particles lose exploration capacity and the swarm collapses toward a common attractor before meaningful Pareto diversity is achieved. Under the multi-objective formulation of (14), this stagnation implies that PSO effectively reduces the three-objective problem to a scalarized proxy, which may not yield balanced trade-offs between acceleration tracking accuracy, modal consistency, and physical regularization.

EANN, by contrast, achieves a competitive central tendency with markedly reduced run-to-run variability, a combination that is directly relevant for control-oriented deployment where repeated identification under varying conditions is required. The balanced trade-off between median accuracy and bootstrap reliability supports the suitability of EANN for downstream controller and observer synthesis.

B. Multi-Objective Quality and Control-Theoretic Implications

The dominance analysis reported in Section V-C reveals a pronounced asymmetry: EANN dominates GA 351,294 times against only 35,410 in the reverse direction. This asymmetry is not captured by the hypervolume indicator alone, since both EANN and GA achieve comparable hypervolume values of 1.0000 and 0.99999, respectively. The critical distinction lies in the internal distribution of Pareto solutions: EANN generates a denser and more uniformly distributed front comprising 98 non-dominated solutions versus 77 for GA, providing finer resolution of trade-off candidates between acceleration tracking accuracy, modal consistency (f_2), and physical regularization (f_3).

From a control-theoretic perspective, this richer Pareto structure has direct consequences for structural model selection. As established by the Bauer–Fike bound in (50), reduced

dispersion in the identified parameters θ directly constrains the eigenvalue uncertainty of the state matrix $A(\theta)$. A denser Pareto front provides the structural engineer with a broader set of candidate models from which to select, allowing simultaneous satisfaction of accuracy requirements and minimization of parametric uncertainty, rather than accepting a single compromise solution imposed by premature convergence. In adaptive control and digital twin scenarios, this flexibility is fundamental: different operating regimes may prioritize either time-domain tracking fidelity or modal consistency, and the richer Pareto structure accommodates both.

C. Parametric Robustness and Physical Plausibility

The coefficient of variation analysis reported in Section V-D quantifies the practical consequences of the multi-objective behavior described above. GA produces CV values exceeding 2.5 and 4.0 for the damping coefficients c_3 and c_4 , respectively, indicating that the identified energy dissipation properties of intermediate floors are effectively unreliable across stochastic runs. Under the state-space formulation of (9), such variability directly propagates into uncertainty in the predicted modal damping ratios, which in turn affects controller tuning margins, observer convergence dynamics, and the predictability of the closed-loop structural response.

PSO presents near-zero CV for all parameters, but as established in Section V-C, this apparent stability reflects stagnation rather than physical robustness. Because PSO collapses to a single solution per run, the near-zero CV is a consequence of premature convergence and does not carry the same physical interpretation as low dispersion achieved through genuine multi-objective exploration.

EANN maintains CV values within 0.07–0.10 across all floors and damping coefficients, and produces significantly smoother spatial parameter profiles, as quantified by the smoothness index (52). These profiles better reflect the underlying physical continuity of the structure: since all columns are fabricated from the same material, stiffness and damping are expected to vary monotonically along the building height. Such physical coherence reduces the risk of numerically accurate but structurally unrealistic parameter distributions, which can compromise model validity in model predictive control and observer-based state estimation applications. These findings are consistent with recent evidence showing that physics-guided neural models improve robustness and physical interpretability in structural identification problems [42].

D. Computational Trade-Off and Deployment Feasibility

As analyzed theoretically in Section IV and confirmed experimentally in Section V-F, the EANN framework incurs a higher offline training cost (438.75 ± 1.01 s per run) compared with GA (91.53 ± 14.93 s) and PSO (51.34 ± 6.82 s). However, this comparison must be interpreted within the correct operational context.

For GA and PSO, the reported runtime corresponds to the cost incurred at *every new identification instance*: both methods require solving the full multi-objective optimization problem expressed in (14) for each new dataset. Accordingly,

their online identification cost scales as $C_{\text{reopt}} \approx P \cdot G \cdot C_{\text{sim}}$, as formalized in (38). For EANN, the training cost is incurred only once offline; subsequent identification reduces to a deterministic forward pass with complexity $\mathcal{O}(d_\theta)$, as derived in (37), executable in milliseconds. In structural health monitoring and adaptive control scenarios where parameters must be updated repeatedly as new measurements arrive, this distinction is fundamental and constitutes the primary practical advantage of the proposed framework over classical reoptimization-based approaches.

E. Limitations and Future Directions

The present investigation is restricted to a linear five-degree-of-freedom benchmark structure subjected to a single excitation profile derived from the 1985 Mexico City earthquake record. The extent to which the observed trends generalize to highly nonlinear, large-scale, or strongly coupled structures remains an open question. In particular, hysteretic behavior, magnetorheological damper nonlinearities, and time-varying structural properties introduce identification challenges that lie beyond the scope of the linear mass–spring–damper formulation adopted here. Furthermore, the observability of damping parameters under accelerometer-only sensing is inherently limited, and broadband stochastic excitation profiles may expose additional sensitivity not captured by the single seismic record used.

Future work will address these limitations along three directions: extending the physics-guided EANN to nonlinear building models equipped with magnetorheological dampers for semi-active vibration control; incorporating heterogeneous sensing modalities including strain gauges and velocity sensors to improve parameter observability; and developing on-line adaptive identification variants capable of tracking time-varying stiffness and damping properties following damage events or structural modifications. Additionally, surrogate-assisted evolutionary strategies and transfer learning across structural configurations offer promising avenues for reducing the offline training burden under structural degradation scenarios.

VII. CONCLUSION

This paper has presented and experimentally validated a physics-informed multi-objective Evolutionary Artificial Neural Network (EANN) framework for accelerometer-only structural parameter identification of a five-story building prototype under seismic base excitation. The framework addresses a fundamental limitation of classical metaheuristic identification approaches: the need to solve a new global optimization problem for each identification instance. By decoupling the computationally intensive training phase from deployment, the proposed paradigm reduces structural parameter estimation to a deterministic forward pass after a single offline evolutionary search, enabling near real-time integration into structural health monitoring pipelines and adaptive control architectures. The experimental validation was conducted over 30 independent stochastic runs per method and benchmarked against GA and PSO baselines under identical conditions.

The experimental evidence establishes three principal findings. First, from an *accuracy* standpoint, EANN achieves a 5.0% reduction in median RMSE and a 9.8% reduction in derived median MSE relative to PSO, with statistically significant differences confirmed by Kruskal–Wallis and post-hoc Wilcoxon rank-sum tests ($p < 0.001$) across all metrics. Nonparametric bootstrap confidence intervals further confirm that EANN produces finite and comparatively narrow bounds across all primary indicators, indicating improved statistical stability under stochastic optimization conditions.

Second, from a *robustness* standpoint, EANN maintains a coefficient of variation within 0.07–0.10 for damping coefficients across all floors, in sharp contrast with GA, which produces CV values exceeding 2.5 and 4.0 in intermediate floors. Under the Bauer–Fike theorem (50), this reduced parametric dispersion translates directly into tighter eigenvalue uncertainty bounds on the state matrix $A(\theta)$, enhancing observer convergence reliability and robustness margins against modeling mismatch. The smoother spatial parameter profiles produced by EANN additionally reflect the physical continuity expected from a uniformly fabricated structure, improving compatibility with model-based control and structural health monitoring applications.

Third, from a *multi-objective* standpoint, EANN generates a denser and more uniformly distributed Pareto front comprising 98 non-dominated solutions, dominating GA 351,294 times against only 35,410 in the reverse direction. This dominance asymmetry, which is not captured by the hypervolume indicator alone, confirms that EANN provides finer resolution of trade-off candidates between tracking accuracy, modal consistency, and physical regularization — a property that directly facilitates model selection for subsequent controller and observer design.

Taken together, these findings demonstrate that the practical usability of a structural identification framework depends not only on its median accuracy but on the stability of its parametric estimates, the physical coherence of its spatial parameter profiles, and its suitability for repeated deployment within control and monitoring loops. The proposed EANN framework simultaneously advances all three dimensions, establishing a methodological foundation for physics-informed, statistically reliable structural identification under accelerometer-only sensing conditions and providing a deployable inference mechanism that is compatible with real-time structural health monitoring, digital twin updating, and adaptive vibration control.

ACKNOWLEDGMENT

The authors are grateful to M. Sc. Rolando Carrera for helping us to complete the experiments at the Vibration Control Laboratory at the Instituto de Ingeniería, UNAM and Eng. Fernando Maldonado Salgado for the support on the software and the high performance computer.

Declaration of interest: None.

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VIII. BIOGRAPHY SECTION



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His professional and research activities have included academic and technological consulting, participation in national and international research projects, and supervision of undergraduate and graduate students. He has been recognized as a national peer evaluator by the Ministry of Science, Technology and Innovation (Minciencias) in areas such as automation and control systems, artificial intelligence, robotics, telecommunications, and electronic engineering.

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